

DATA VISUALIZATION PROJECT

DUE SUNDAY 10/11 AT 11:59PM

CSCI/MATH 608 FALL 2026

OVERVIEW

You will demonstrate your ability to import, clean, and visualize a dataset of your choosing.

SKILLS TESTED

- Importing datasets with Pandas
- Cleaning datasets with Pandas
- Formulating data science questions
- Visualizing datasets with Altair
- Explaining visualizations

STEP ONE: CHOOSE A DATASET AND RESEARCH QUESTIONS

Choose a dataset that you are interested in exploring. It could be the dataset you used in the Data Collection Project, or a different dataset.

Basic challenge: select a tabular dataset

Moderate challenge: select a graph/network dataset

Advanced challenge: select a text corpus

Come up with three research questions relating to your dataset that can be answered by visualizations (i.e., descriptive and exploratory questions).

STEP TWO: IMPORT AND CLEAN YOUR DATA

Create a new .ipynb file to contain your work for the project. Add your research questions to the top of the file. Start by importing your data, and then wrangle it into a tidy format to answer your research questions. Be sure to include all of the code you used to import and clean the data within the .ipynb file. You should reference the slides/lecture for help with these tasks. If cleaning your data requires code that we did not cover, I recommend you reach out to me and/or reference the [Pandas User Guide](#). AI-generated code is permitted only for data cleaning tasks that we did not cover in class, and I highly recommend taking the time to understand what the code is doing and not simply copying and pasting.

STEP THREE: CREATE AND EXPLAIN YOUR VISUALIZATIONS

For each of the three research questions you came up with, create a visualization that answers the question as completely and concisely as possible. Follow the guidelines for clearly conveying your message that were given in the lecture. Include all of the code for the visualization in your .ipynb file. If creating your visualizations requires code that we did not cover, I recommend you reach out to me and/or reference the [Altair User Guide](#). AI-generated code is permitted only for data visualization tasks that we did not cover in

class, and I highly recommend taking the time to understand what the code is doing and not simply copying and pasting.

For each of your visualizations, include in your .ipynb file a text cell thoroughly explaining the visualization based on the criteria specified in lecture.

STEP FOUR: REVIEW AND SUBMIT YOUR WORK

Review the grading rubric and make any necessary changes, then upload your .ipynb file to Canvas.

GRADING RUBRIC

	Master (100%)	Adept (83.33%)	Apprentice (66.67%)	Novice (50%)
Importing Data (10%)	Dataset is imported in an appropriate way	Dataset is imported with a few mistakes	Dataset is imported, but incorrectly	No dataset is imported
Cleaning Data (15%)	Data is cleaned appropriately for all visualization analyses conducted (i.e., in a tidy format), and all code used for cleaning is included in the submission	Data is cleaned appropriately for 2 of the 3 visualization analyses conducted and/or some code used for cleaning is not included in the submission	Data is cleaned appropriately for 1 of the 3 visualization analyses conducted, some of the code used for cleaning may not be included in the submission	Data is not cleaned appropriately or no code used for cleaning is included in the submission
Data Science Questions (15%)	Three questions are included in the submission that are appropriate to be answered by visualizations (i.e., descriptive or exploratory questions)	Two questions are included in the submission that are appropriate to be answered by visualizations (i.e., descriptive or exploratory questions)	One question is included in the submission that is appropriate to be answered by visualizations (i.e., descriptive or exploratory questions)	No questions are included in the submission that are appropriate to be answered by visualizations (i.e., descriptive or exploratory questions)

Visualizations (30%)	All three visualizations answer a given research question simply and plainly, following the guidelines for clearing conveying your message, and all code creating these visualizations is included in the submission	All three visualizations answer a given research, with some small areas for improvement regarding simplicity and adhering to the guidelines for clearing conveying your message; all code creating these visualizations is included in the submission	All three visualizations approach answering a given research, with some sizable areas for improvement regarding simplicity and adhering to the guidelines for clearing conveying your message and/or some code used to create the visualizations is missing	The visualizations fail to answer any of the research questions or are missing
Explanations (30%)	All three visualizations are clearly explained, following the guidelines for visualization explanations.	All three visualizations are explained, with some small room for improvement regarding following the guidelines for visualization explanations.	All three visualizations are explained, with sizable room for improvement regarding following the guidelines for visualization explanations.	One or more visualization explanations are missing